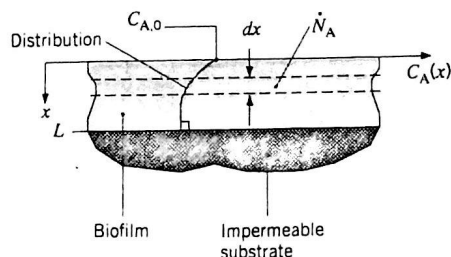


87

COURSE: CHE 312: Transport Phenomena II**Fall Semester 2025****Department of Chemical Engineering****University of Illinois Chicago****Midsemester Exam #4****Only one sheet of paper 8 1/2 x 11 is allowed****No phones or i-pads or computers are allowed****Duration: 75 minutes --- December 4, 2025****All Problems must be done****Problem 1: (35 points)**

Biofilms, which are colonies of bacteria that can cling to living or inert surfaces, can cause a wide array of human infections. Infections caused by bacteria living within biofilms are often chronic because antibiotics that are applied to the surface of a biofilm have difficulty penetrating through the film thickness. Consider a biofilm that is associated with a skin infection. An antibiotic (species A) is applied to the top layer of a biofilm (species B) so that a fixed concentration of medication, $C_{A,0} = 4 \times 10^{-3} \text{ kmol/m}^3$, exists at the upper surface of the biofilm. The diffusion coefficient of the medication within the biofilm is $D_{AB} = 2 \times 10^{-12} \text{ m}^2/\text{s}$. The antibiotic is consumed by biochemical reactions within the film, and the consumption rate depends on the local concentration of medication expressed as $\dot{N}_A = -k_1 C_A$ where $k_1 = 0.1 \text{ s}^{-1}$. To eradicate the bacteria, the antibiotic must be consumed at a rate of at least $0.2 \times 10^{-3} \text{ kmol/s} \cdot \text{m}^3$ ($\dot{N}_A \leq -0.2 \times 10^{-3} \text{ kmol/s} \cdot \text{m}^3$) since, at smaller absolute consumption rates, the bacteria will be able to grow back faster than it is destroyed. Determine the maximum thickness of a biofilm, L , that may be treated successfully by the antibiotic.

Schematic:

We have, $0 = D_{AB} \frac{d^2 C_A}{du^2} - k_1 C_A$

$$\frac{d^2 C_A}{du^2} - \frac{k_1}{D_{AB}} C_A = 0 \quad \left(\text{Let } \lambda^2 = \frac{k_1}{D_{AB}} \right)$$

$$\frac{d^2 C_A}{du^2} - \lambda^2 C_A = 0$$

We have the equation $x^2 - \lambda^2 = 0 \quad \therefore x = \pm \lambda$

$$\therefore C_A(u) = A e^{\lambda u} + B e^{-\lambda u}$$

or $C_A(u) = A \cosh(\lambda u) + B \sinh(\lambda u)$

Boundary conditions

BC1: $C_A(0) = C_{A,0}$

at $u=0$, $\cosh(0) = 1$ & $\sinh(0) = 0$

~~BC~~ $\therefore C_A(0) = A = C_{A,0}$

BC2: no flux at $u = L_f$

$$\left. \frac{dC_A}{du} \right|_{u=L_f} = 0$$

$$\therefore \frac{dC_A}{du} = A \lambda \sinh(\lambda u) + B \lambda \cosh(\lambda u)$$

at $u = L_f$

$$0 = A \lambda \sinh(\lambda L_f) + B \lambda \cosh(\lambda L_f)$$

$$A \sinh(\lambda L_6) + B \cosh(\lambda L_6) = 0$$

$$B = \frac{-A \sinh(\lambda L_6)}{\cosh(\lambda L_6)} = \frac{-C_{A,0} \sinh(\lambda L_6)}{\cosh(\lambda L_6)}$$

For seaweed minimum concentration for effective killing

$$N_A = -k_2 C_A C_n$$

$$N_A \leq 0.2 \times 10^{-3} \text{ kmol/m}^3$$

$$\therefore k_2 C_A C_n \leq 0.2 \times 10^{-3}$$

To be effective $C_A(L_6) \geq 2 \times 10^{-3}$

or we can write (substituting $C_A(L_6)$ using previous eq)

$$\frac{C_{A,0}}{\cosh(\lambda L_6)} \geq 2 \times 10^{-3}$$

$$\frac{4 \times 10^{-3}}{\cosh(\lambda L_6)} \geq 2 \times 10^{-3}$$

or $\cosh(\lambda L_6) \leq 2$

$$\lambda = \sqrt{\frac{0.1}{2 \times 10^{-12}}} = 2.236 \times 10^5$$

$$\lambda L_6 \leq \cosh^{-1}(2)$$

using identity $\cosh^{-1}(u) = \ln(u + \sqrt{u^2 - 1})$

$$\cosh^{-1}(2) = \ln(2 + \sqrt{2^2 - 1}) = 1.317$$

Therefore

$$|L_6| \leq \frac{1.317}{2.236 \times 10^3} \Rightarrow \leq 5.86 \times 10^{-6}$$

Problem 2: (35 points)

The efficacy of pharmaceutical products is reduced by prolonged exposure to high temperature, light, and humidity. For water vapor-sensitive consumer products that are in tablet or capsule form, and might be stored in humid environments such as bathroom medicine cabinets, *blister packaging* is used to limit the direct exposure of the medicine to humid conditions until immediately before its ingestion.

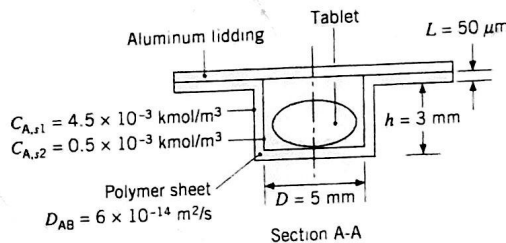
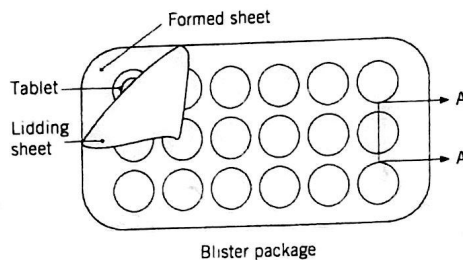
Consider tablets that are contained in a blister package composed of a flat *lidding sheet* and a second, *formed sheet* that includes troughs to hold each tablet. The formed sheet is $L = 50 \mu\text{m}$ thick and is fabricated of a polymer material. Each trough is of diameter $D = 5 \text{ mm}$ and depth $h = 3 \text{ mm}$. The lidding sheet is fabricated of aluminum foil. The binary diffusion coefficient for water vapor in the polymer is $D_{AB} = 6 \times 10^{-14} \text{ m}^2/\text{s}$ while the aluminum may be assumed to be impermeable to water vapor. For molar concentrations of water vapor in the polymer at the outer and inner surfaces of $C_{A,1} = 4.5 \times 10^{-3} \text{ kmol/m}^3$ and $C_{A,2} = 0.5 \times 10^{-3} \text{ kmol/m}^3$, respectively, determine the rate at which water vapor is transferred through the trough wall to the tablet.

$$L = 50 \mu\text{m} = 5 \times 10^{-5} \text{ m}$$

$$D = 5 \text{ mm} = 5 \times 10^{-3} \text{ m}$$

$$h = 3 \text{ mm} = 3 \times 10^{-3} \text{ m}$$

Schematic:



No generation

1D steady Diffusion

D_{AB} is constant

$$\frac{d}{dn} \left(D_{AB} \frac{dC_A}{dn} \right) = 0$$

$$D_{AB} \frac{d^2 C_A}{dn^2} = 0$$

$$\frac{d^2 C_A}{dn^2} = 0$$

$$\frac{dC_A}{dn} = C_1$$

$$C_A(n) = C_1 n + C_2$$

using Boundary conditions

at $n=0$

$$C_A(0) = C_{A,1} \Rightarrow C_2 = C_{A,1}$$

at $n=L$

$$C_A(L) = C_{A,2} \Rightarrow C_1 L + C_{A,1} = C_{A,2}$$

$$\therefore C_1 = \frac{C_{A,2} - C_{A,1}}{L}$$

$$C_A(x) = C_{A,1} + \frac{C_{A,2} - C_{A,1}}{L} x$$

$$C_{A,1} - C_{A,2} = 4.0 \times 10^{-3} \text{ kmol/m}^3$$

$$N_A = \frac{D_{AB} (C_{A,1} - C_{A,2})}{L} = \frac{(6 \times 10^{-14}) (4 \times 10^{-3})}{5 \times 10^{-5}} \\ = 4.8 \times 10^{-12} \text{ kmol/(m}^2\text{s)}$$

$$A = A_{\text{catten}} + A_{\text{side}}$$

$$A_{\text{catten}} = \pi (2.5 \times 10^{-3})^2 = \pi (6.25 \times 10^{-6}) = 1.9635 \times 10^{-5} \text{ m}^2$$

$$A_{\text{side}} = \pi (5 \times 10^{-3}) (3.0 \times 10^{-3}) = 4.7124 \times 10^{-5} \text{ m}^2$$

$$A = 6.6759 \times 10^{-5}$$

$$N_A = \cancel{\text{something}} N_A * A = (4.8 \times 10^{-12}) (6.6759 \times 10^{-5}) \\ = 3.20 \times 10^{-16} \text{ kmol/s}$$

Problem 3: (35 points)

A plane wall is a composite of two materials, A and B. The wall of material A has uniform heat generation $\dot{q} = 1.5 \times 10^6 \text{ W/m}^3$, $k_A = 75 \text{ W/m} \cdot \text{K}$, and thickness $L_A = 50 \text{ mm}$. The wall material B has no generation with $k_B = 150 \text{ W/m} \cdot \text{K}$ and thickness $L_B = 20 \text{ mm}$. The inner surface of material A is well insulated, while the outer surface of material B is cooled by a water stream with $T_\infty = 30^\circ\text{C}$ and $h = 1000 \text{ W/m}^2 \cdot \text{K}$.

1. Sketch the temperature distribution that exists in the composite under steady-state conditions.
2. Determine the temperature T_0 of the insulated surface and the temperature T_2 of the cooled surface.

② $L_A = 50 \text{ mm} = 0.05 \text{ m}$, $L_B = 20 \text{ mm} = 0.02 \text{ m}$

B.c

Inner surface of A is insulated

$$\frac{dT}{dn} = 0 \quad , \quad -k_B \left. \frac{dT}{dn} \right|_{n=L_A+L_B} = h(T_2 - T_\infty)$$

$n=0$, Material A = $0 \leq n \leq L_A$
 at insulated surface
 Material B = $L_A \leq n \leq L_A + L_B$

Region A: $\frac{d^2T}{dn^2} = -\frac{\dot{q}}{k_A} \Rightarrow \frac{dT}{dn} = -\frac{\dot{q}}{k_A} n + C_1$

at insulated boundary, therefore, $C_1 = 0$, so $\frac{dT}{dn} = -\frac{\dot{q}}{k_A} n$

$$T(n) = -\frac{\dot{q}}{2k_A} n^2 + C_2$$

at $n=0$, $T(0) = T_0 = C_2$

$$\therefore T(n) = -\frac{\dot{q}}{2k_A} n^2 + T_0$$

at $n = L_A \Rightarrow T(L_A) = T_0 - \frac{\dot{q}(L_A)^2}{2k_A} \Rightarrow \Delta T_A = \frac{1.5 \times 10^6 (0.05)^2}{2(75)} = 25^\circ\text{C}$
 (To conduct)

Region B: $\frac{d^2 T}{dn^2} = 0 \Rightarrow \frac{dT}{dn} = C_3 \Rightarrow T(n) = C_3 n + C_4$

at $n = L_A$ & at $n = L_A + L_B$

$T(L_A) = T_1$ $-k_B \frac{dT}{dn} = h(T_2 - T_\infty)$

$\therefore -k_B C_3 = h(T_2 - T_\infty)$

$T_2 = T(L_A + L_B) = C_3(L_A + L_B) + C_4$

$T_1 - T_2 = -C_3 L_B$

at $n = L_A \Rightarrow q'' = -k_A \left. \frac{dT}{dn} \right|_{L_A}$

$\therefore q'' = k_A \left(\frac{-\dot{q}}{k_A} L_A \right)$ (from $\frac{dT}{dn} = \frac{-q''}{k_A} n$)

$q'' = \dot{q} L_A$

$\therefore q'' = 1.5 \times 10^4 (0.02) = 75 \times 10^4 \text{ W/m}^2$

$\therefore$ at convection surface

$q'' = h(T_2 - T_\infty) \Rightarrow \therefore T_2 = 75 + 30 = 105^\circ\text{C}$

$\therefore T_1 = T_2 + \frac{q'' L_B}{k_B}$

$T_1 = 105 + \frac{75 \times 10^4 \times 0.02}{150}$
 $= 115^\circ\text{C}$

$T_0 = 115 + 25 = 140^\circ\text{C}$

① Temperature drop through material A would be parabolic which through material B would be linear